

**CLASSIFICATION OF ELEMENTS
AND PERIODICITY IN PROPERTIES***NCERT-Based Topic-wise MCQ*

1. Periodic classification of elements is used to examine

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- (a) periodic trends in physical properties of elements
- (b) periodic trends in chemical properties of elements
- (c) both (a) and (b)
- (d) none of the above

2. Which of the following is a set containing man-made elements?

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- (a) Gold, Mercury, Zinc
- (b) Gold, Radium, Palladium
- (c) Curium, Osmium, Uranium
- (d) Technetium, Neptunium, Einsteinium

3. * Cl, Br and I form a Dobereiner's triad. If the atomic masses of Cl and I are 35.5 and 127, respectively, the atomic mass of Br is

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- (a) 162.5
- (b) 91.5

(c) 81.25

(d) 45.625

4. If two members of a Dobereiner triad are phosphorus and antimony, the third member of this triad is

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- (a) arsenic
- (b) sulphur
- (c) iodine
- (d) calcium

5. * Lothar Meyer plotted physical properties such as atomic volume, melting point and ...X... against atomic weight. Here, X refers to

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- (a) mass
- (b) boiling point
- (c) surface tension
- (d) none of these

6. * The most significant contribution towards the development of the periodic table was made by

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- (a) Mendeleev
- (b) Avogadro
- (c) Dalton



(d) Cavendish

7. Who developed the long form of the periodic table?

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- (a) Lothar Meyer
- (b) Niels Bohr
- (c) Mendeleev
- (d) Moseley

8. Which fact is not valid for Dobereiner's triads?

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- (a) The atomic weight of the middle element is roughly the average of the other two elements.
- (b) The properties of the middle element are roughly the average of the other two elements.
- (c) The elements of triads belong to the same group of the modern periodic table.
- (d) The elements of triads have the same number of valence electrons.

9. * Select the correct chronological order for the discoveries of the following scientists: Dobereiner, Newlands, Chancourtois and Mendeleev.

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- (a) Chancourtois, Dobereiner, Newlands, Mendeleev
- (b) Dobereiner, Chancourtois, Newlands, Mendeleev
- (c) Dobereiner, Newlands, Chancourtois, Mendeleev
- (d) Chancourtois, Newlands, Dobereiner, Mendeleev

10. * The molecular formulae of the chlorides of Eka-Aluminium and Eka-Silicon, respectively, are

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- (a) GaCl_3 and SiO_4
- (b) GaCl_3 and AlCl_3
- (c) AlCl_3 and SiCl_4
- (d) GaCl_3 and GeCl_4

11. In Mendeleev's periodic table, which of the following elements, instead of having a lower atomic weight, was placed after an element of higher atomic weight, thereby ignoring the order of increasing atomic weights?

- (a) Iodine



(b) Antimony

(c) Bromine

(d) Molybdenum

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(a) 18

(b) 8

(c) 30

(d) 32

12. Which of the following is correct about Eka-Aluminium and Eka-Silicon?

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(a) The oxide of Eka-Aluminium is Ga_2O_3 and the oxide of Eka-Silicon is GeO_2 .

(b) The oxide of Eka-Aluminium is Al_2O_3 and the oxide of Eka-Silicon is Si_2O_3 .

(c) The melting point of Eka-Aluminium is lower than the melting point of Eka-Silicon.

(d) Both (a) and (c)

15. The long form of the periodic table consists of

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(a) seven periods and eight groups

(b) seven periods and eighteen groups

(c) eight periods and eighteen groups

(d) eighteen periods and eight groups

13. Which of the scientists given below discovered that the periodic table should be based on atomic number?

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(a) Mendeleev

(b) Newlands

(c) Moseley

(d) Lothar Meyer

16. Modern periodic table is based on the atomic number of the elements. The experiment that proved the significance of atomic number was

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(a) Millikan's oil-drop experiment

(b) Moseley's work on X-ray spectra

(c) Bragg's work on X-ray diffraction

(d) Discovery of X-rays by Röntgen

14. How many elements are there in the 6th period of the periodic table?



17. * The symbol and IUPAC name for the element with atomic number 120, respectively, are

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- (a) Ubn and unbnilium
- (b) Ubn and unbiunium
- (c) Ubn and unnibium
- (d) Ubn and unnilium

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- (a) 5th
- (b) 15th
- (c) 3rd
- (d) 17th

18. What is the IUPAC name of the element with atomic number 114?

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- (a) Unununnium
- (b) Unnilquadium
- (c) Ununquadium
- (d) Unnilennium

19. * The IUPAC nomenclature of an element with electronic configuration $[Rn] 5f^{14} 6d^1 7s^2$ is

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- (a) Unnilbium
- (b) Unnilunium
- (c) Unnilquadium
- (d) Unniltrium

20. * An element with electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^3$ belongs to which group of the periodic table?

